UNIT – 2	9 Hours
Analysis of DC Circuits	
Analysis of DC multi loop circuit for both independent and dependent sources: mesh, node, super mesh and super node.	

Introduction

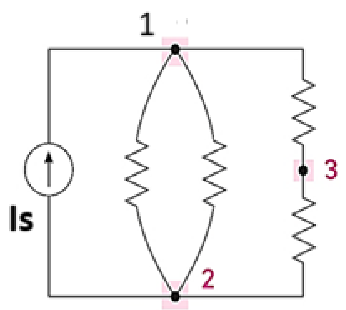
The interconnection of various electric elements in a prescribed manner known as an electric circuit to perform a desired function. The electric elements include controlled and uncontrolled source of energy, resistors, capacitors, inductors, etc. Analysis of electric circuits refers to computations required to determine the unknown quantities such as voltage, current and power associated with one or more elements in the circuit. To contribute to the solution of engineering problems one must acquire the basic knowledge of electric circuit analysis and laws.

DC Circuits

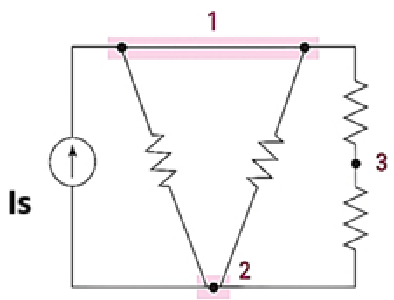
A closed circuit in which dc current flows is called a dc circuit.

NODES and LOOPS in Electric Circuit

Node: point at which two or more elements have a common connection is called a **node.** For example, Fig. 2.1a shows a circuit containing three nodes.



(a)



(b)

Fig.2.1 (a) A circuit containing three nodes and five branches (b) Node 1 is redrawn to look like two nodes: it is still one node.

Node 1 in Fig. 2.1a, can be shown as two separate junctions connected by a (zero-resistance) conductor, as in Fig. 2.1b.

Loop: In an electric circuit, a loop is a closed path formed by connecting circuit elements in a way that current can flow through them continuously without any breaks. If the node at which we started is the same as the node on which we end, then this is closed path or a **loop**. Consider the Fig 2.2,

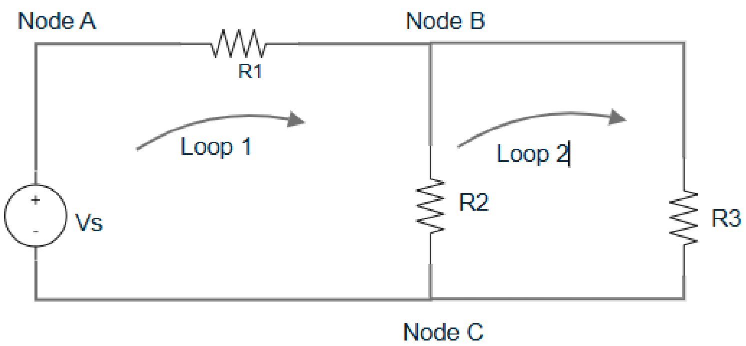


Fig 2.2 Two Loop electric circuit.

There are two loops:

(i) Loop1: A-B-C-A

(ii) Loop2:B-C-B

Basic Laws of Electrical Engineering:

Ohms Law: Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all physical conditions and temperatures remain constant.

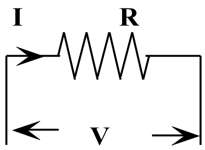


Fig 2.3 Ohm's law example

$$\mathbf{I = \frac{V}{R}} \qquad \textbf{(2.1)}$$

KIRCHOFF LAWS

Gustav Kirchhoff, a German physicist, observed that these were instances of two general conditions fundamental to the analysis of any electrical network. These conditions may be stated as follows:

KIRCHOFF'S CURRENT LAW

Kirchhoff's Current Law (KCL) states that the total current entering a node (junction) in an electrical circuit is equal to the total current leaving that node. This is also known as Kirchhoff's first law or the junction rule. Essentially, it reflects the principle of conservation of charge, meaning that charge is not lost or gained within a circuit.

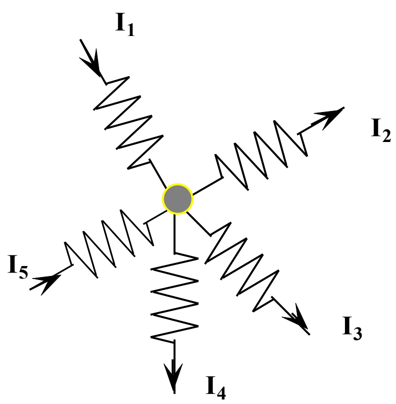


Fig 2.4 Kirchoff's current law example

Convention: Current entering a node -ve, leaving the node +ve

-I1+I2+I3+I4-I5=0

i.e. I2+I3+I4=I1+I5

Exercise 1: Find the magnitude and direction of the unknown currents in Fig 2.5. Given i_1=10A, i_2=6A and i_5=5A

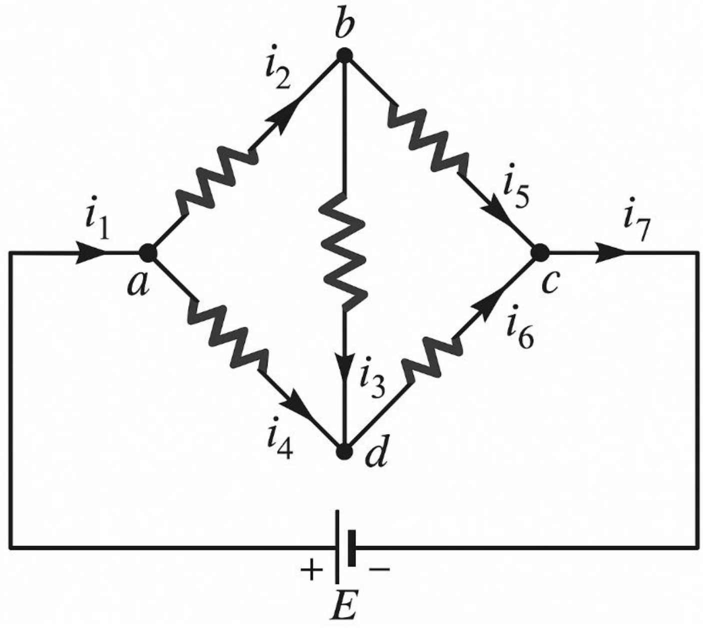


Fig.2.5 Exercise 1

By Applying KCL $i_1 = i_7$ i.e. $i_7 = 10A$

At node "a" $i_1 = i_2 + i_4$

$10 = 6 + i_4$, hence $i_4 = 4A$

At node "b", Applying KCL

$i_2 = i_3 + i_5$

or $i_3 = i_2 - i_5 = 6 – 5 = 1A$

At node "c" $i_7 = i_5 + i_6$

$i_6 = i_7 - i_5 = 10 – 5 = 5A$

Hence $i_1 = i_7 = 10A$

$i_2 = 6A$

$i_3 = 1A$

$i_4 = 4A$

$i_5 = 4A$

$i_6 = 5A$

Note: The directions of currents can be arbitrarily assigned by applying KCL and can be changed if the obtained values are negative.

Exercise 2: Find **i_1, i_2, i_3 and i_4 shown in Fig.2.6**

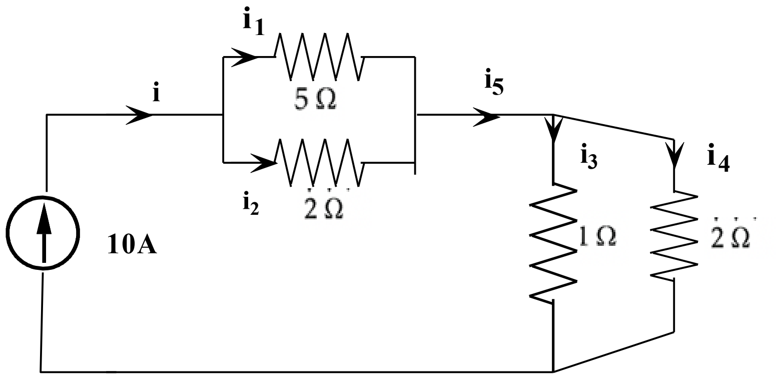


Fig.2.6 Exercise 2

Using the current division method,

$$i_1 = i\,\frac{2}{5+2} = 10\,\frac{2}{5+2} = 2.857\ \text{A}$$

$$i_2 = 10\,\frac{5}{5+2} = 7.143\ \text{A}$$

$$i_3 = i\,\frac{2}{1+2} = 6.67\ \text{A}$$

$$i_4 = i\,\frac{1}{1+2} = 3.33\ \text{A}$$

KIRCHOFF'S VOLTAGE LAW

Kirchhoff's Voltage Law (KVL) states that the sum of all voltage drops around any closed loop in a circuit is equal to zero. This law is a consequence of energy conservation in electrical circuits. In simpler terms, the total voltage source (like a battery) in a loop must equal the total voltage drops across the components in that loop. Consider the closed electric circuit shown in Fig 2.7

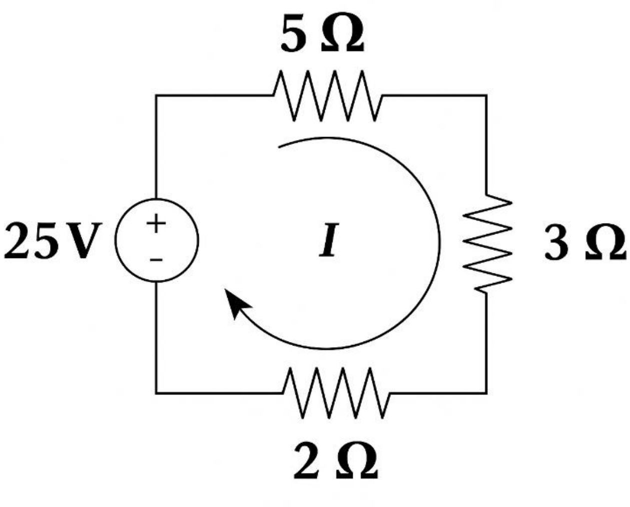


Fig 2.7 Kiechoff's voltage law example

Convention: Voltage drops -ve. Source – to + is + ve , + to – is -ve

Hence applying to KVL to the circuit shown in Fig.2.9

-5I-3I-2I+25=0

10I=25

I=2.5 A

ELECTRIC ENERGY SOURCES:

They are of two types.

Independent Sources

Independent sources are circuit elements that provide a specified voltage or current irrespective of other elements in the circuit. Their output is not controlled by any variable elsewhere in the circuit.

Independent Voltage Source

An independent voltage source is a two-terminal device that provides a constant voltage regardless of the current flowing through it. Representation of Independent Voltage source is shown in Fig 2.8

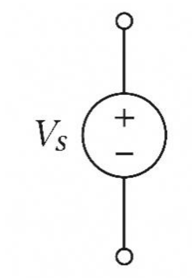


Fig.2.8 Independent voltage source

Examples:

Battery: Supplies a fixed DC voltage like 1.5V or 12V regardless of the current drawn.

DC Generator: Provides a constant voltage used in backup systems or power supplies.

Independent Current Source

An independent current source delivers a constant current regardless of the voltage across its terminals. Its representation is as shown in Fig 2.9

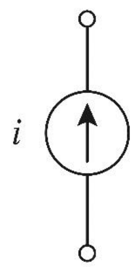


Fig.2.9 Independent current source

Examples:

Current-regulated power supply: Used in applications like LED driving where a fixed current is needed.

Mesh analysis or Loop analysis of electric circuit with Independent Sources

The mesh analysis, the name being derived from the similarities in appearance between the closed loops of the network and a physical “fence” or mesh. The current through a mesh is known as mesh current. In this method, a distinct current is assumed in the loop and the polarities of drops in each element in the loop are determined by the assumed direction of loop current for the loop. KVL is then applied around each closed loop and by solving this loop equation, the branch currents are determined.

The following steps are used in Mesh analysis

1. Identify the meshes in an Electric Circuit. Consider the circuit shown in Fig 2.10

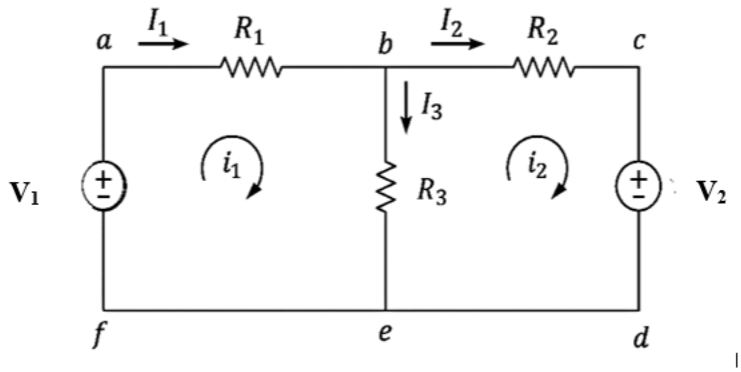


Fig 2.10 Example for Mesh analysis

There are two meshes:

Mesh 1: abefa

Mesh 2: bcdeb

2. Assign the loop currents. Current i_1 flows in loop 1 and current i_2 flows in loop 2.
3. Apply KVL to each loop

   Consider loop1:

   $-I_1R_1-(I_1-I_2)\ R_3+V_1=0$

   $-(R_1+R_3)\ I_1+R_3I_2=-V_1$

   $(R_1+R_3)\ I_1-R_3I_2=V_1$ (2.2)

   Consider loop2:

   $-I_2R_2-V_2-(I_2-I_1)\ R_3=0$

   $R_3I_1-(R_1+R_3)\ I_2=V_2$ (2.3)

   Solve simultaneous equations (2.2) and (2.3) to find I_1 and I_2 using equation solver in the calculator

Exercise 3: For the circuit shown in Fig. 2.11, find the branch currents using mesh analysis.

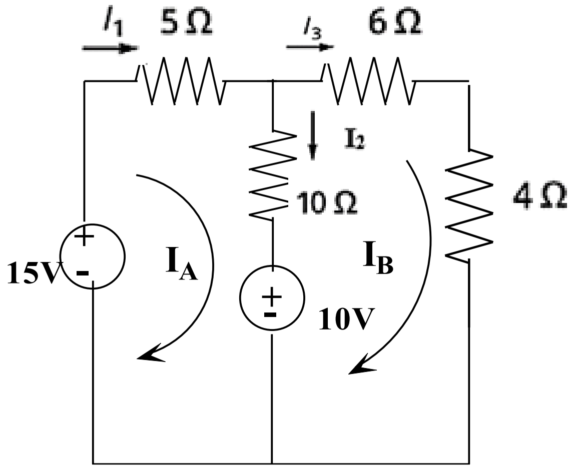


Fig. 2.11 Exercise 3

Write KVL to Mesh 1,

$15 - 5I_A - 10\ (I_A - I_B) - 10 = 0$

$-15I_A + 10I_B = -5$

$15I_A - 10I_B = 5$ (2.4)

Write KVL equations to Mesh 2,

$-6I_B - 4I_B + 10 - 10(I_B - I_A) = 0$

$10I_A - 20I_B = -10$

$-10I_A + 20I_B = 10$ (2.5)

Solving (2.4) and (2.5),

$I_A = 1A\ ,\ I_B = 1A$

$\mathbf{I_1 = 1A}$

$\mathbf{I_2 = 0A}$

$\mathbf{I_3 = 1A}$

Exercise 4: For the circuit shown in Fig.2.12, find the mesh currents using mesh analysis. Also find the power dissipated in each resistor and power supplied/absorbed by each source

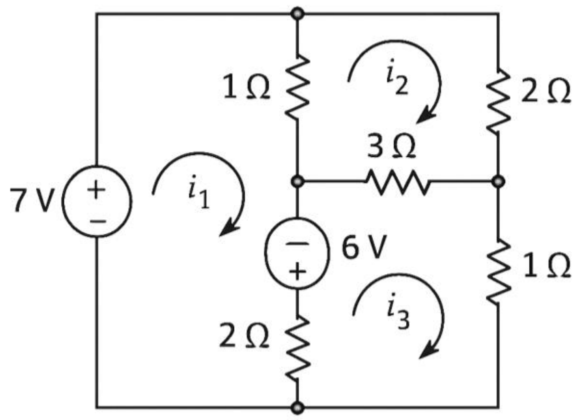


Fig.2.12 Exercise 4

Applying KVL to Mesh 1 (i_1),

$$7 - (i_1 - i_2) + 6 - 2(i_1 - i_3) = 0$$

$$-3i_1 + i_2 + 2i_3 = -13$$

$$3i_1 - i_2 - 2i_3 = 13 \quad (2.6)$$

Applying KVL to Mesh 2 (i_2),

$$-2i_2 - 3(i_2 - i_3) - 1(i_2 - i_1) = 0$$

$$-i_1 + 6i_2 - 3i_3 = 0 \quad (2.7)$$

Applying KVL to Mesh 3 (i_3),

$$-3(i_3 - i_2) - i_3 - 2(i_3 - i_1) - 6 = 0$$

$$2i_1 + 3i_2 - 6i_3 = 6 \quad (2.8)$$

Solving (2.6), (2.7) and (2.8)

$\mathbf{i_1 = 6.69A}$

$\mathbf{i_2 = 2.31A}$

$\mathbf{i_3 = 2.38A}$

Nodal analysis of electric circuit with Independent Sources

Nodal analysis is a method that provides a general procedure for analysing the circuits using node voltages as the circuit variables. It is also called Node-Voltage Method.

In the nodal method, the number of independent node-pair equations needed is one less than the number of junctions in the network. If "n" denotes the number of independent node equations of "j" the number of junctions, then n = j – 1

The following steps are used in Node analysis

1.Identify the nodes in an Electric Circuit. Consider the circuit shown in Fig.2.13

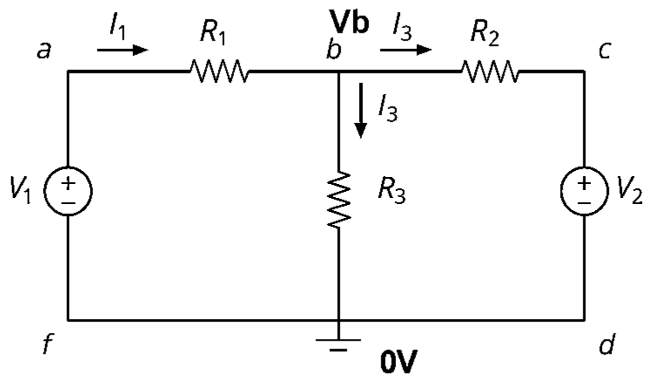


Fig.2.13 Example for Nodal Analysis

There are two nodes:

Node 1: b

Node 2: e

2. Assign the node voltages at different nodes. Voltage V_b at node b and 0V at node e (datum node)
3. Apply KCL at node b

$$\frac{(V_b - V_1 - 0)}{R_1} + \frac{(V_b - 0)}{R_3} + \frac{(V_b - V_2 - 0)}{R_2} = 0$$

4. Solve for V_b using equation solver in the calculator

Compared with the mesh method (where KVL equations are frequently used to solve the network), the nodal methos is advantageous when the network has many parallel circuits; otherwise, both the nodal and mesh methods offer almost equal advantages for the mesh method the number of independent mesh equations needed is $m = b - (j - 1)$, where b is the number of branches.

Note: If m < n the mesh method is preferred and m > n, nodal method is preferred.

Exercise 5: Using Nodal method, find the current through r_2 shown in Fig.2.14

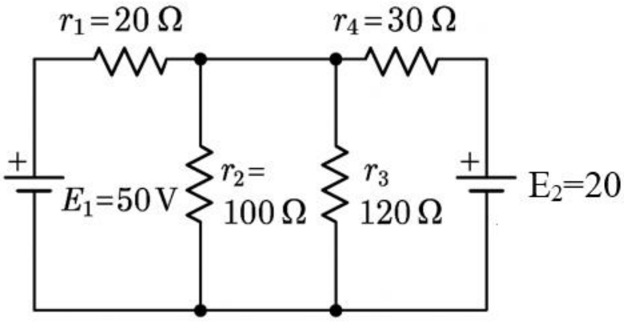


Fig.2.14 Exercise 5

Solution: The circuit can be redrawn and written as shown in Fig.2.15 with naming the nodes and branch currents.

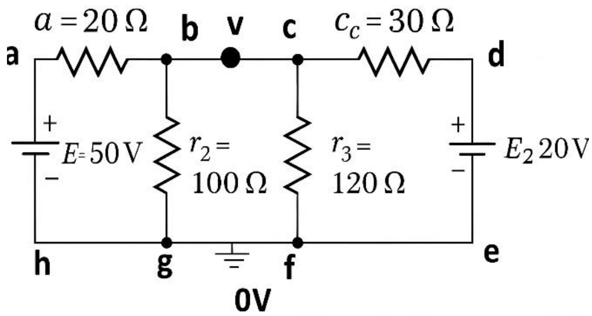


Fig.2.15

From the Figure, $V_c = V_b = v$ (Node b and c are at the same potential)

Applying KCL at node b, $\frac{(v-50)}{20} + \frac{(v-0)}{100} + \frac{(v-0)}{120} + \frac{v-20}{30} = 0$ (2.9)

Solving the equation (2.9) for v , v = 31.18 V

$$i_2 = \frac{v}{r_2} = \frac{31.18}{100} \; 0.3118\,A$$

The current through r_2 = 311.8 mA

Exercise 6: Using nodal method, find the currents supplied/absorbed by the batteries in the circuit shown in Fig.2.16

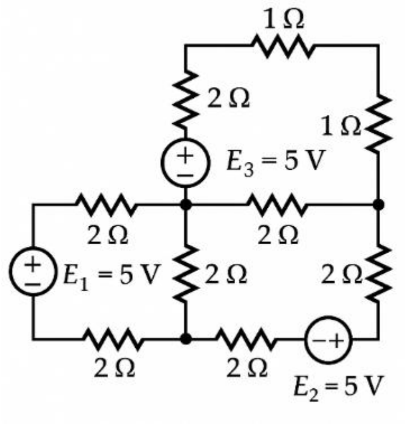


Fig.2.16 Exercise 6

By redesignating the given circuit with nodal names and respective potentials as shown in Fig.2.17

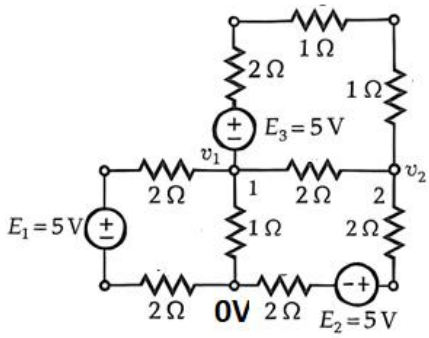


Fig.2.17

At node (1)

$$\frac{v_1 - 5 - 0}{4} + \frac{v_1 - 0}{1} + \frac{v_1 + 5 - v_2}{4} + \frac{v_1 - v_2}{2} = 0$$

$2v_1 - 0.75v_2 = 0$ (2.10)

At node (2)

$$\frac{v_2 - 5 - 0}{4} + \frac{v_2 - v_1}{2} + \frac{v_2 - 5 - v_1}{4} = 0$$

$-0.75v_1 + v_2 -= 2.5$ (2.11)

Solving (2.10) and (2.11)

$\mathbf{v_1 = 1.3V}$ **and** $\mathbf{v_2 = 3.47\ V}$

Current supplied by $E_1 = \frac{0+5-v_1}{4} = 0.925$ A

Current supplied by $E_2 = \frac{0+5-v_2}{4} = 2.225$ A

Current supplied by $E_3 = \frac{v_1+5-v_2}{4} = 0.706$ A

Exercise 7: In the circuit shown in Fig.2.18 Find the current through 1Ω resistor.

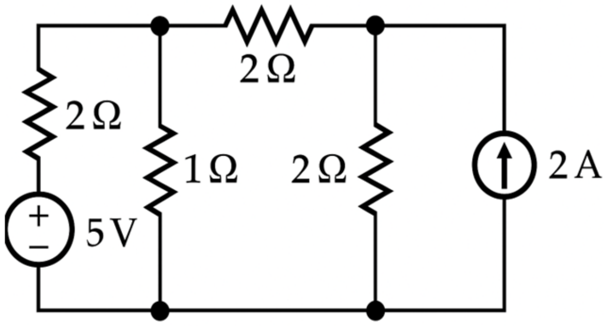
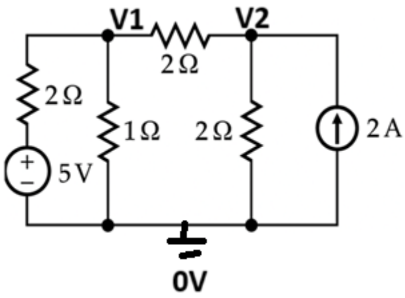


Fig.2.18 Exercise 7



Let the node voltages be V1 and V2

Using Nodal Analysis

At node 1; $\frac{V_1-5-0}{2}+\frac{V_1-0}{1}+\frac{V_1-V_2}{2}=0$

$V_1\left(\frac{1}{2}+\frac{1}{1}+\frac{1}{2}\right)-\frac{1}{2}V_2=2.5$

$2V_1-0.5V_2=2.5$ (2.12)

At node 2; $-2+\frac{V_2-0}{2}+\frac{V_2-V_1}{2}=0$

$V_2\left(\frac{1}{2}+\frac{1}{2}\right)-\frac{1}{2}V_1=2$

$-0.5V_1+V_2=2$ (2.13)

Solving equations (2.12) and (2.13) ,

$V_1=2$ V

Hence the current through 1Ω resistor is $\frac{V_1}{1}=\mathbf{2\ A}$

DEPENDENT SOURCES

Dependent sources, also known as controlled sources, are circuit elements whose output (voltage or current) is determined by a voltage or current in another part of the circuit. Examples: Amplifiers, Transistors, Operational Amplifiers.

Dependent or controlled sources are of the following types:

Voltage Controlled Voltage Source (VCVS):

A VCVS is a type of dependent source where the output voltage is directly proportional to a controlling voltage elsewhere in the circuit.

The Symbol and representation is as shown in Fig.2.19

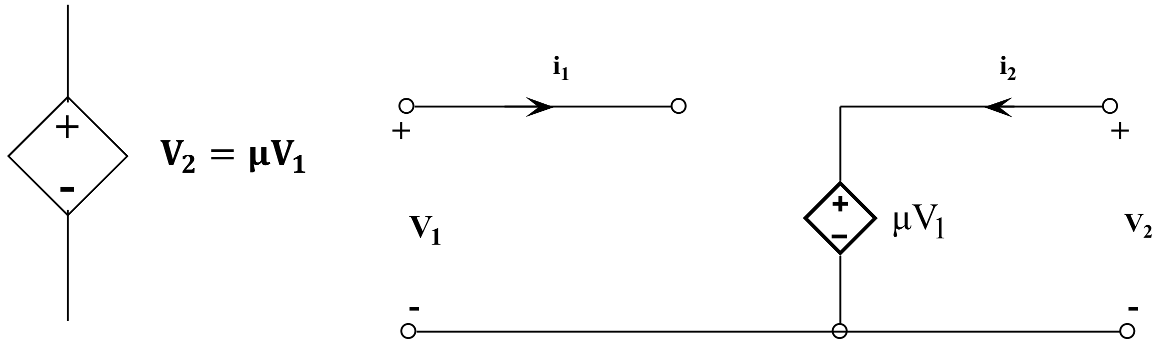


Fig.2.19 Voltage controlled voltage source

Examples:

Opamp: An ideal operational amplifier (op-amp) used as an amplifier. In this case, the output voltage is controlled by the input voltage, with the amplification factor determined by the op-amp's gain.

Oscillators: where they help control the frequency of oscillation by controlling the output voltage.

Filters: Reject certain frequencies.

Current Controlled Voltage Source (CCVS)

A current-controlled voltage source (CCVS) is a type of dependent voltage source whose output voltage is proportional to a current elsewhere in the circuit. The proportionality constant is called the trans resistance, typically denoted by r.

The Symbol and representation is as shown in Fig.2.20

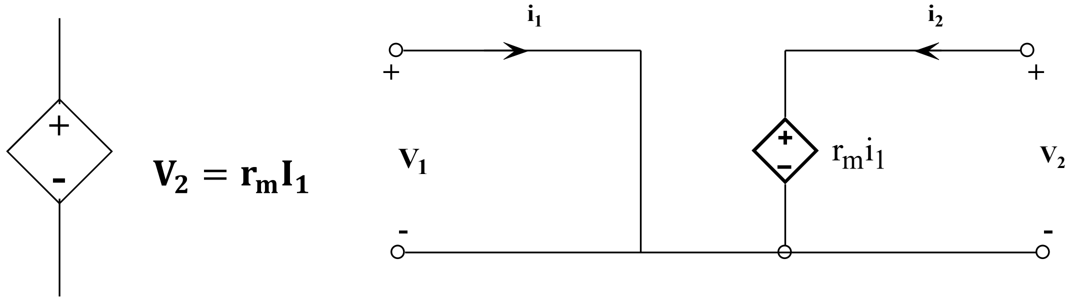


Fig.2.20 Current controlled voltage source

Examples:

Bipolar Junction Transistor (BJT): The output voltage (V_{CE}) is controlled by the input current (I_B).

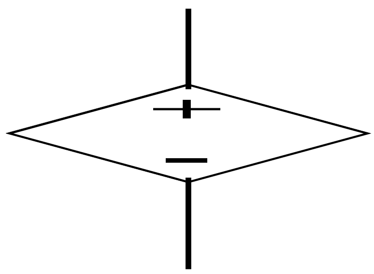


Fig.2.21 BJT Circuits

By controlling base current I_B, Voltage between collector and emitter V_{CE} can be varied.

Voltage Controlled Current Source (VCCS)

A voltage-controlled current source (VCCS) is a circuit element where the output current is directly proportional to a control voltage.

The Symbol and representation is as shown in Fig.2.22

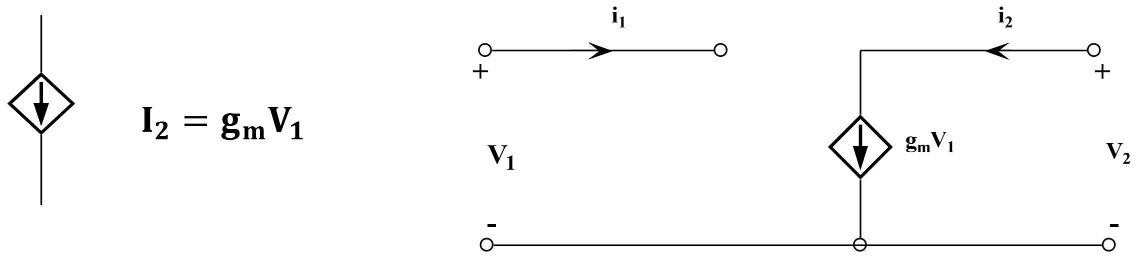


Fig.2.22 Voltage controlled current source

Example: Field Effect Transistor (FET): Gate voltage controls the drain current.

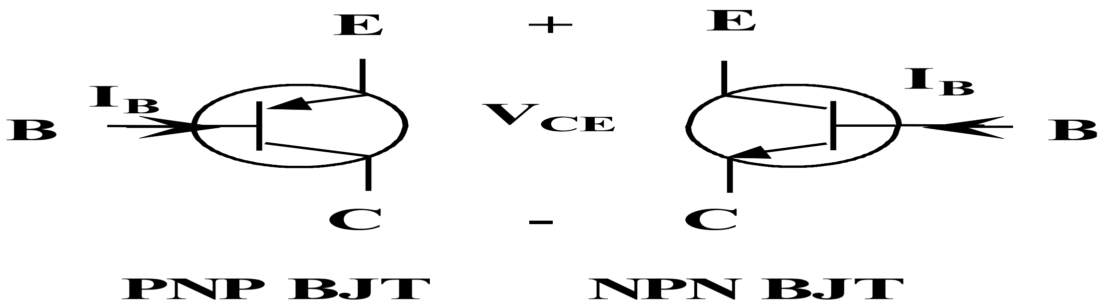


Fig.2.23 FET circuit

Metal Oxide semiconductor field effect transistor (MOSFET) small signal model is shown in Fig. 2.23. g_m is called transconductance of the transistor and is given by $\frac{I_D}{V_{GS}}$.One can also find the applications of Voltage control current sources in digital-to-analog converter circuits and voltage-to-current converter circuits.

Current Controlled Current Source (CCCS)

A Current-Controlled Current Source (CCCS) is a current source whose output current is controlled by a current in another part of the circuit. Essentially, the CCCS acts like a current amplifier, where a small input current can control a larger output current.

The Symbol and representation is as shown in Fig.2.24

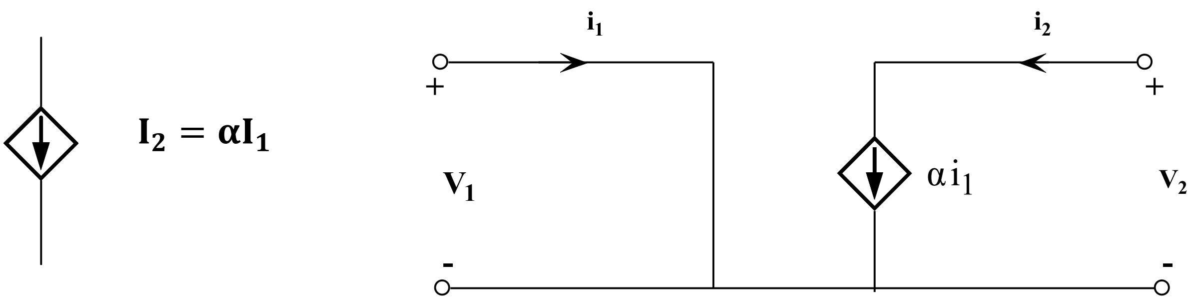


Fig.2.24 Current controlled voltage source

Examples: BJT operated in a common-emitter configuration. The collector current is controlled by the base current, making it a CCCS.

An operational amplifier (Op-amp) can also be used as a CCCS by connecting a resistor in the feedback loop. The output current through the resistor is controlled by the input current.

Mesh analysis or Loop analysis of electric circuit with dependent Sources

Mesh analysis with dependent sources involves applying the same principles as with independent sources, but with an additional step to account for the dependency.

Steps for Mesh Analysis with Dependent Sources:

1. Identify the meshes and define Mesh Currents:

Assign a current direction (usually clockwise) to each mesh in the circuit.

2. Write KVL Equations:

Apply Kirchhoff's Voltage Law (KVL) to each mesh. This step is similar to mesh analysis with independent sources. Pay close attention to voltage polarities across resistors and sources.

3. Identify Dependent Sources:

Locate the dependent sources within the circuit.

4. Form Constraint Equations:

For each dependent source, create an equation that relates the dependent source's value to other currents or voltages in the circuit. These equations are based on the type of dependent source (voltage-controlled voltage source, current-controlled current source, etc.).

5. Solve the Equations: Solve the equations to find the mesh currents using equation solver in Calculator

Exercise 8: Find v_x in the Fig.2.25. What is the power absorbed by the dependent source?

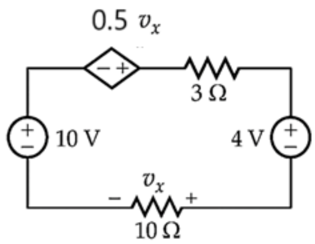
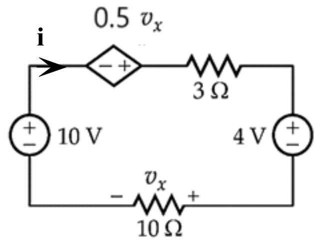


Fig.2.25 Exercise 8



Applying KVL to the loop,

$$10 + 0.5V_x - i(3 + 10) - 4 = 0$$

$13i = 6 + 0.5V_x$ (2.14)

However,

$V_x = i \times 10\Omega$ (2.15)

Substituting (2.14) in (2.15)

$13i = 6 + 0.5 \times 10i$

$13i = 6 + 5 I$

$8i = 6$

$i = 0.75$ A

From (2),

$V_x = 0.75 \times 10 = 7.5V$

The current enters the dependent source from the -ve terminal and exits from the +ve terminal. Hence the dependent source supplies power.

The voltage of the dependent source is

$0.5V_x = 0.5 \times 7.5 = 3.75V$

P $_{\text{Dep.Source}}$ = 3.75 V x 0.75 A = 2.81W

Exercise 9

Using Mesh analysis, find the value of **α** for the circuit shown in Fig. 2.26 when the power loss in 1Ω resistor is 9W.

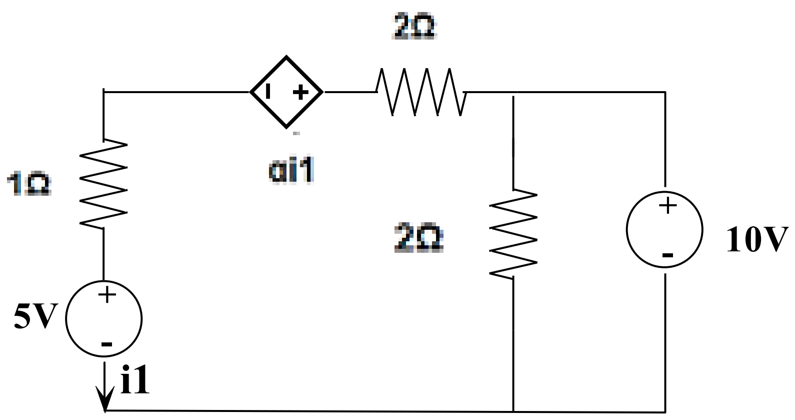
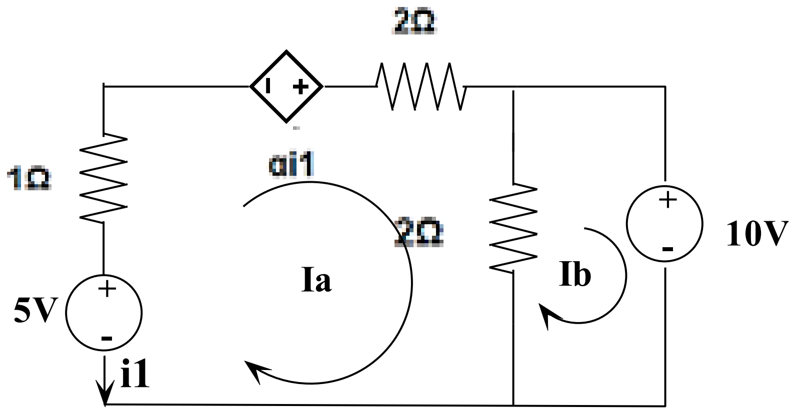


Fig.2.26 Exercise 9

Solution:



$i_1^2 x\, 1 = 9$ or $i_1 = 3A = Ia$

Consider the first loop,

$\alpha i_1 - 2i_1 - 2(i_1 - I_b) + 5 - 1i_1 = 0$

$(\alpha - 2 - 2 - 1)i_1 + 2I_b = -5$

$(\alpha - 5)x3 + 2I_b = -5$ (2.16)

Consider the second loop,

$-10 - 2(I_b - i_1) = 0$

$-10 - 2I_b + 2x3 = 0$

$-2I_b - 4 = 0$

$I_b = -2A$ (2.17)

Substituting the value of I_b in equation (2.16),

$(\alpha - 5)x3 + 2x - 2 = -5$

$(\alpha - 5)x3 = -1$

$\alpha = \frac{-1+15}{3} = \frac{14}{3} = 4.67$

Nodal analysis of electric circuits with dependent Sources

Nodal analysis with dependent sources involves applying the same principles as with independent sources, but with an additional step to account for the dependency. Apply Kirchhoff's Current Law (KCL) at each non-reference node, expressing currents in terms of node voltages and dependent source parameters.

Steps involved are:

1. Identify all the nodes:

Locate all nodes in the circuit. Choose a reference node (ground) and assign voltage variables to the remaining nodes.

2. Identify Dependent Sources:

Determine the type and location of any dependent sources (voltage-controlled voltage source (VCVS), current-controlled current source (CCCS), voltage-controlled current source (VCCS), or current-controlled voltage source (CCVS)).

3. Write KCL Equations:

Apply KCL at each non-reference node. Express all currents in terms of node voltages and the dependent source parameters.

4. Incorporate Dependent Source Equations:

For dependent sources, write equations that relate the controlling voltage/current to the controlled voltage/current.

5. Solve the System of Equations:

Sove the system of linear equations obtained by earlier steps to obtain the node voltages

Exercise 10:

In the network shown in the Fig.2.27, Find v_1, v_2 and v_3 by nodal analysis.

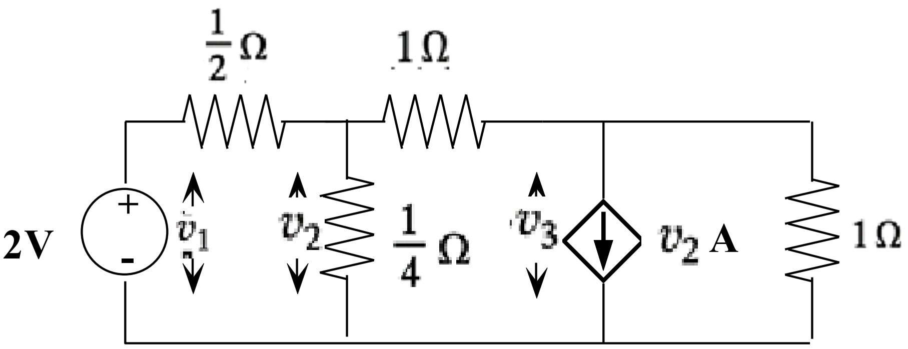
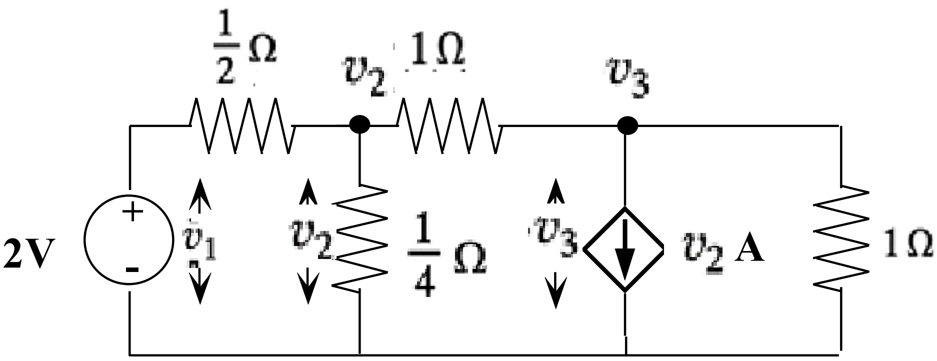


Fig.2.27 Exercise 10



By inspection, it is evident that $v_1 = 2V$

At node 2

$$\frac{v_2 - 2}{\frac{1}{2}} + \frac{v_2}{\frac{1}{4}} + \frac{v_2 - v_3}{1} = 0$$

$$7v_2 - v_3 = 4 \tag{2.18}$$

At node 3,

$$\frac{v_3 - v_2}{1} + v_2 + \frac{v_3}{1} = 0$$

$2v_3 = 0$ (2.19)

Thus,

$v_2 = {}^4\!/_7$ V

Exercise 11: In Fig 2.28, using node analysis, find the current in the resistor between node a and b.

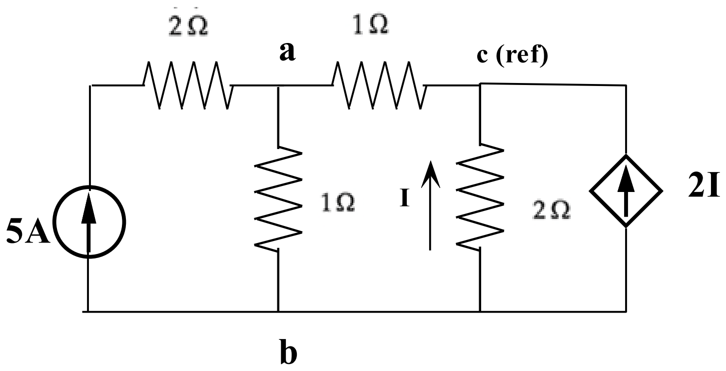


Fig 2.28 Exercise 11

At node a, Nodal analysis gives, $-5 + \frac{v_a - v_b}{1} + \frac{v_a}{1} = 0$

$2v_a - v_b = 5$ (2.20)

At node b, Nodal analysis gives, $\frac{v_b - v_a}{1} + 5 + 2I + \frac{v_b}{2} = 0$

But $I = \frac{v_b}{2}$

$$\frac{v_b - v_a}{1} + 5 + 2\frac{v_b}{2} + \frac{v_b}{2} = 0$$

$-v_a + \frac{5}{2}v_b = -7$

$2v_a - 5v_b = 14$ (2.21)

Solving (2.20) and (2.21),

We get $v_a = 1.375V$ and $v_b = -2.25V$

The 1 Ω resistor between node a and b,

$$I_{ab} = \frac{v_a - v_b}{1}$$

$$= \frac{1.375 + 2.25}{1}$$

$$= \mathbf{3.625A}$$

Super Mesh Analysis:

A super mesh is a special technique used in mesh analysis when a current source is located in a branch that is shared by two adjacent meshes. To solve the circuit, the two meshes are combined into a single, larger loop. You then apply Kirchhoff's Voltage Law (KVL) around this super mesh, and a separate auxiliary equation is created to relate the two mesh currents to the value of the current source.

Steps involved in solving electric circuits with super mesh are as follows

1. Identify the meshes in an Electric Circuit. Assign loop currents
2. Identify Super meshes: Locate current sources that are present in the closed loops of a given Electric Circuit.
3. Assign the super mesh current values as the current source values present in the closed loop
4. Apply KVL: Write KVL equations for each remaining individual meshes that are not part of a super mesh.
5. Solve the Equations using the equation solver in the calculator

Exercise 12: Find the current 'i' in Fig.2.29 by mesh analysis

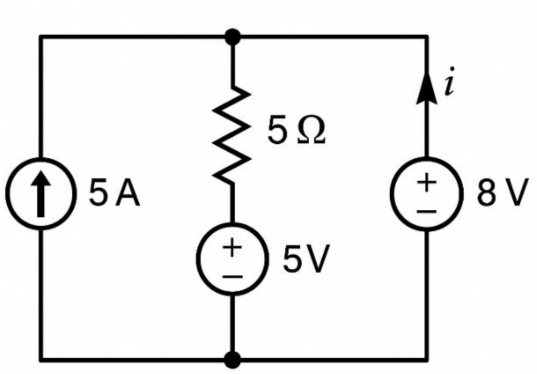


Fig.2.29 Exercise 12

Redraw the circuit assigning the nodes and currents as shown in Fig.2.30

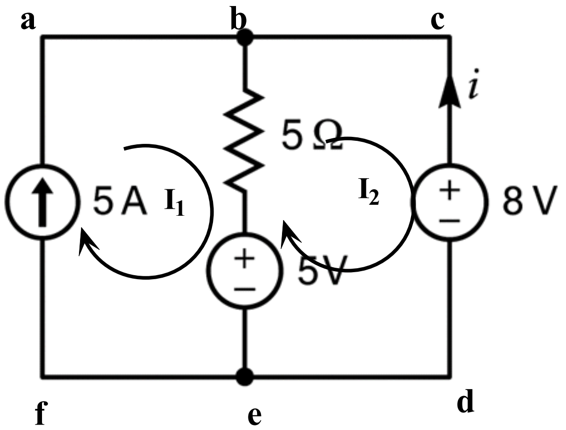


Fig.2.30

Solution:

There is 5A current source in the mesh abefa.

Therefore, I_1=5 A.

Applying KVL to mesh bcdeb, We get:

$-8+5-5(I_2- I_1)=0$

$-3-5\ I_2+ 5I_1=0$

$5\ I_2+3=5\ I_1$

Substitute **I_1=5 A**:

I_2=4.4 A

Exercise 13: Determine i_1 for the circuit shown in Fig .2.31 by mesh analysis

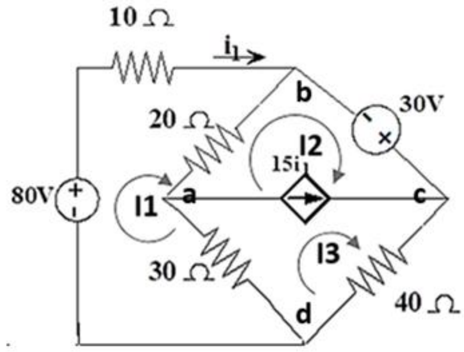


Fig .2.31 Exercise 13

From the Circuit, it can be inferred that $I_1=i_1$

There is a dependent current source of $15i_1$,

$I_3 - I_2 = 15i_1 = 15I_1$

$15I_1+I_2-I_3=0$ (2.22)

Applying KVL, to the mesh bcdab:

$-20 (I_2-I_1)+30 - 40I_3 - 30(I_3 - I_1) = 0$

$50I_1 -20I_2 -70 I_3 = -30$ (2.23)

Applying KVL to first mesh badb:

$+80 - 10I_1- 20 (I_1-I_2) - 30 (I_1 - I_3)= 0$

$-60I_1 +20I_2+30I_3=-80$

$60I_1 -20I_2-30I_3=80$ (2.24)

Solving equations (2.22) in (2.23) and (2.24),

I_1=0.58A

Super Node Analysis:

Whenever a voltage source (Independent or Dependent) is connected between the two nodes without any resistance in series with it, then these two nodes form a generalized node called the Super node. So, Super node can be regarded as a surface enclosing the voltage source and its two terminals.

Steps involved in solving electrical circuits with super node are as follows: -

1. Identify total number of nodes
2. One node selected as reference node and it is assigned to have ground(zero) potential and remaining nodes called as non-reference nodes and we assign voltage designations to non-reference nodes.
3. Check for super node and write equation
4. Develop KCL equations for each non reference nodes
5. Solve the equation to find the unknown node voltages

Exercise 14: Find the voltages V_1 and V_2 for the circuit shown in Fig.2.32 by node analysis.

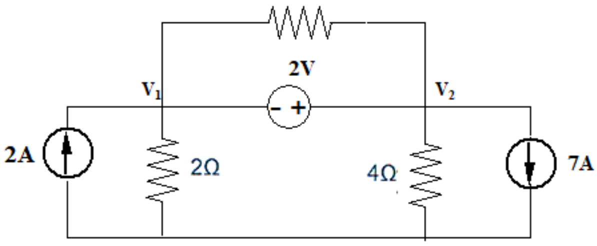


Fig.2.32 Exercise 14

Solution: Here 2V voltage source is connected between Node-1 and Node-2 and it forms a Super node. Hence $v_2 - v_1 = 2$ V

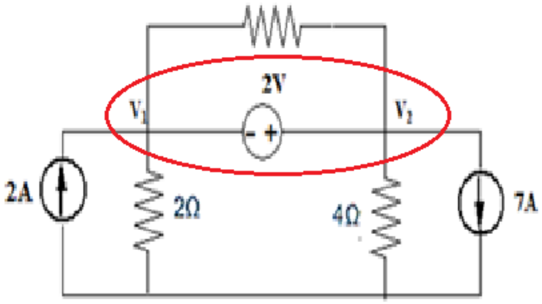


Fig.2.32

2V source between the nodes 1 and 2,

$v_1+2-v_2=0$

$v_2-v_1=2$ (2.25)

Applying KCL at this super node, $-2+\frac{V_1}{2}+7+\frac{V_2}{4}=0$

$$\frac{v_1}{2}+5+\frac{v_2}{4}=0$$

$2v_1 + v_2 = -20$ (2.26)

Solving equations (2.25) and (2.26) ,

v_1 = **-7.333V**

v_2 = -5.333v

Exercise 15: Find the node voltages V_1 and V2 for the circuit shown below in Fig.2.33.

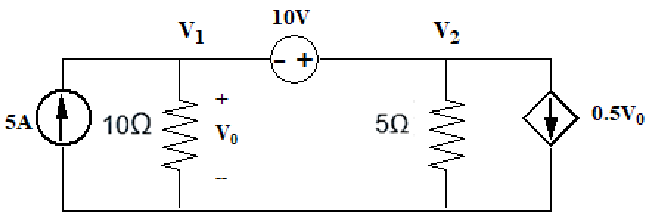
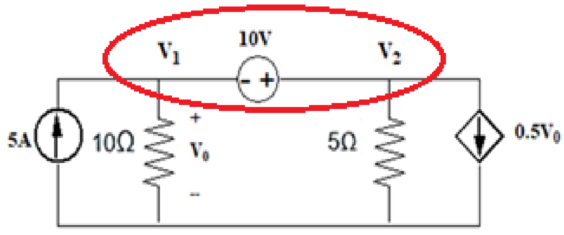


Fig.2.33 Exercise 15



In the figure we see that 10V source is connected between the nodes 1 and 2, it forms a super node.

$V_1 + 10 - V_2 = 0$

$V_2 - V_1 = 10$ (2.27)

Therefore, applying KCL at the super node, we get

$$-5 + \frac{V_1}{10} + \frac{V_2}{5} + 0.5V_0 = 0$$

$-5 + 0.1V_1 + 0.2V_2 + 0.5V_0 = 0$

But $V_1 = V_0$ Therefore, the equation becomes

$-5 + 0.1V_1 + 0.2V_2 + 0.5V_1 = 0$

$0.6V_1 + 0.2V_2 = 5$ (2.28)

Solving equations (2.27) and (2.28),

$V_1 = 3.75V$ and $V_2 = 13.75V$